Martin
Poinsinet de Sivry-Houle

PhD in Engineering Physics

Quantitative Software, Numerical Computing, and Python/C++

Markelerbergpad 971
Amsterdam, Netherlands
+31085293413
martin.poinsinet.de.sivry@gmail.com
MartinPdeS

Professional summary

PhD-level computational physicist and scientific software developer with strong Python and C/C++ experience, numerical methods expertise, and a focus on performance-oriented systems. Builds research software that turns mathematical models into tested, reusable tools, including a C++-backed Python trading/backtesting package and end-to-end simulation pipelines for noisy measurement data. Comfortable working across research and engineering concerns: algorithm design, validation, CI/CD, multiprocessing, cross-platform support, and translating exploratory models into maintainable infrastructure.

Education

2015 – 2018 **Master's in Engineering Physics**, *Polytechnic Montreal*

- Focused on optics and photonics with applied programming for data analysis, instrumentation, and numerical workflows.
- Developed tools for smart-camera characterization, optical testing, and measurement automation.
- Contributed to production code in a commercial context during an internship at Matrox Electronic Systems Ltée.

2012 – 2015 **Bachelor of Science in Physics**, *University of Montreal*

- Specialized in physics with a focus on particle physics.
- Completed an internship at CERN involving data analysis in a large research collaboration.

Research & professional experience

2024 – Present **Postdoctoral Researcher in Biomedical Optics (Computational Focus)**, *Amsterdam UMC*, Amsterdam, Netherlands

- Leading development of simulation systems integrating physical models, electronics, signal processing, and software for complex measurement workflows.
- Building first-principles and numerical models, then validating them against experimental behavior through reproducible analysis pipelines.
- Designing model functionality and quantitative metrics for algorithm evaluation, performance analysis, and robust interpretation of noisy time-series data.
- Developing and training CNN models for single-particle detection and classification in high-throughput time-series measurements.
- Prototyping real-time triggering and processing logic on FPGA for low-latency acquisition and hardware-oriented testing workflows.
- Developing practical engineering tools for rapid exploration of design and process tradeoffs, including throughput, robustness, and deployment constraints.
- Maintaining open-source scientific software (PyMieSim, FlowCyPy), including testing, CI/CD, documentation, packaging, and cross-platform usability.

2018 – 2024 **PhD Researcher in Engineering Physics**, *Polytechnic Montreal*, Montreal, Canada

- Designed and built experimental and computational workflows for microscopy and Optical Coherence Tomography (OCT).
- Developed and validated FDTD, EME, and CMT simulation tools for electromagnetic wave propagation and optical component modeling, translating Maxwell's equations into discrete computational formulations and benchmarking results against analytical solutions.
- Implemented light-scattering simulations based on Lorenz-Mie theory and applied Fourier-optics concepts in imaging and optical diagnostics workflows.
- Developed data and image-processing pipelines for experimental measurements, calibration, model validation, and analysis.
- Designed basic electronic circuits for optical sensing and acquisition, supporting integration with measurement hardware.

2016 **Intern**, *Matrox Electronic Systems Ltée*, Dollard-des-Ormeaux, Canada

- Developed an automated protocol for industrial-camera characterization, optical testing, and measurement repeatability analysis in an industrial imaging environment.

2014 **Intern**, *CERN*, Geneva, Switzerland

- Characterized components of the ALICE Inner Tracking System and supported detector testing, experimental integration, and data analysis.

2013 **Intern**, *University of Montreal*, Montreal, Canada

- Contributed to the PICO dark-matter detection experiment through setup, measurements, data analysis, and quantitative validation workflows.

Computing skills

Programming

- Python and C/C++: advanced; Python/C++ interfaces and compiled numerical backends
- Matlab: competent
- LaTeX and Bash: proficient

Diagnostics & analysis

- Optical-model and measurement-analysis workflows with calibration, quality checks, and traceable outputs
- Optical instrumentation support: measurement setup, validation, troubleshooting, and repeatability
- Statistics, uncertainty quantification, risk/performance metrics, and error propagation for quantitative analysis

Simulation & software

- Electromagnetic and computational-optics modeling (FDTD, coupled-mode modeling, Lorenz-Mie, Fourier-optics concepts)
- End-to-end simulation and analysis code for quantitative measurements (signal processing, noise models, reproducible pipelines)
- Linux-based scientific computing, multiprocessing, and performance-oriented implementation
- Git, Docker, CI/CD, testing, and documentation: advanced
- Linux and macOS: advanced; Windows: competent

Extracurricular activities

2022 **President of the student chapter of the SPIE association**

- Management of various activities within the physics department (e.g. conferences, trainings).
- Implication in educational activities for children (e.g. Eurêka festival, school teachings).

2021 **President of the student chapter of the Optical Society of America (OSA)**

- Management of various activities within the physics department (e.g. conferences, trainings).
- Implication in educational activities for children (e.g. Eurêka festival, school teachings).

2019 **Vice president of the student chapter of the Optical Society of America (OSA)**
- Management of various activities within the physics department (e.g. conferences, trainings).
- Implication in educational activities for children (e.g. Eurêka festival, school teachings).

2018 **Reviewer for *Fundamentals of Biophotonic* book from Pr. Boudoux**
- Working on the second edition of the book.
- Review of first edition, adding new chapters on: Laser, diffuse optics, optical design, imaging systems.
- Writing of new exercices and solutions.

Selected publications & scientific outputs

2025 Journal article — Three-mode photonic lanterns: comprehensive analysis from theory to experiments [manuscript]

2024 Journal article — SuPyMode: open-source design and optimization of fiber optic components [manuscript]

2023 Journal article — PyMieSim: fast and flexible far-field Mie scattering simulations [manuscript]

2023 Proceedings article — Extensive Mie scattering analysis for biomedical application using PyMieSim [manuscript]

2021 Journal article — All-fiber few-mode optical coherence tomography using a photonic lantern [manuscript]

2020 Journal article — Coregistered OCT and frequency-encoded multispectral imaging [manuscript]

2025 Patent — Illumination device for microscopy [patent]

2021 Patent — All-fiber multimodal interferometer for imaging technique [patent]

Teaching Experiences

2025 Lab Responsible — Biophotonics
Vrije Universiteit Amsterdam – Coordinated laboratory sessions and practical teaching activities in biophotonics.

2019, 2020, 2022 Lecturer — GBM8802: Biophotonics
Polytechnique Montreal – Delivered graduate-level lectures on optical methods for medicine and biology, including core biophotonics concepts and advanced applications.

2019–2023 Project Manager — PHS1903: Initial Project in Physics Engineering
Polytechnique Montreal – Supervised undergraduate engineering projects, mentoring students in project management, experimental design, and technical problem solving.

2019 Lab Responsible — PHS8210: Fundamentals of Photonics
Polytechnique Montreal – Coordinated laboratory sessions and teaching activities in photonics.

2017–2018 Teaching Assistant — GBM8802: Biophotonics; PHS1903: Initial Project in Physics Engineering; GBM3805: Modern Physics for Biomedical Engineering
Polytechnique Montreal – Supported lectures, grading, laboratory instruction, and student supervision.

Prizes & Scholarships

May 2026 — President's Award for Excellence, ISAC

May 2023 — SUSSP 79 Photonic Sensing & Spectroscopy Summer School Scholarship

May 2022 — Silver Laureate for Community Engagement

March 2021 — FRQNT Research Scholarship

January 2021 — Excellence Scholarship for Graduate Studies

September 2020 — Canadian Research Funding, Mitacs

March 2020 — Excellence Scholarship for Graduate Studies from Foundation J.A. De Sève

February 2020 — SPIE Student Travel Grant

April 2019 Nominee at the Meritas Gala for Best Teacher in Biomedical Engineering at Polytechnique Montréal

September 2018 Doctoral Entrance Scholarship from Polytechnique Montréal

September 2018 Best Poster Presentation at SPIE/OSA École Polytechnique Chapter

May 2014 CERN Summer School Scholarships

Open-source optics & diagnostics software

Open-source software

- **TradeTide** – C++-backed Python trading and backtesting package for strategy prototyping, portfolio simulation, risk management, and performance evaluation; includes multiprocessing, cross-platform support, CI/CD, and GPU-oriented workflows. [Repository]
- **FlowCyPy** – End-to-end simulation framework combining optics, fluidics, electronics, and signal processing for complex measurement systems. [Repository]
- **RosettaX** – Interactive scientific application for data processing, visualization, and analysis workflows. [Repository]
- **PyMieSim** – Python/C++/Fortran package for physics-based light-scattering simulation and particle characterization. [Repository]
- **SuPyMode** – Python/C++ toolkit for numerical simulation and coupled-mode analysis in optical systems. [Repository]
- **PyFinitDiff** – Finite-difference toolbox for engineering and physics simulations. [Repository]
- **FiberFusing** – Python package for numerical modeling of thermal and optical aspects of fiber-fusion processes. [Repository]

Miscellaneous

Languages **French**: native; **English**: fluent; **Spanish**: basic communication; **Dutch**: basic communication.

References

Prof. Caroline Boudoux, caroline.boudoux@polymtl.ca (Research director, Canada)

Prof. Nicolas Godbout, nicolas.godbout@polymtl.ca (Research co-director, Canada)

Dr. Stephane Virally, stephane.virally@polymtl.ca (Research associate, Canada)

Dr. Xavier Attend, x.attendu@amsterdamumc.nl (Researcher, Netherlands)